AS · Cambridge (CIE) · Physics

32 mins 32 questions

Multiple Choice Questions

# Equations of Motion

Displacement, Velocity & Acceleration / Motion Graphs / Area under a Velocity-Time Graph / Gradient of a Displacement-Time Graph / Gradient of a Velocity-Time Graph / Deriving Kinematic Equations / Solving Problems with Kinematic Equations / Acceleration of Free Fall Experiment / Projectile Motion

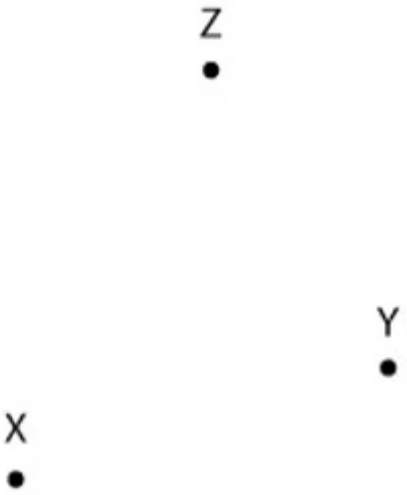
| | |
|---|---|
| Easy (11 questions) | /11 |
| Medium (11 questions) | /11 |
| Hard (10 questions) | /10 |
| **Total Marks** | **/32** |

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# Easy Questions

**1** An object moves directly from X to Z




In a shorter time, a second object moves from X to Y to Z.

Which statement about the two objects is correct for the journey from X to Z?

**A.** they have the same average speed

**B.** they have the same average velocity

**C.** they have the same displacement

**D.** they travel the same distance

**(1 mark)**

**2** An experiment is done to measure the acceleration of free fall of a body from rest.

Which measurements are needed?

**A.** the height of fall and the time of fall

**B.** the height of fall and the weight of the body

**C.** the mass of the body and the height of fall

**D.** the mass of the body and the time of fall


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**(1 mark)**

**3** Which feature of a graph allows acceleration to be determined?

**A.** the area under a displacement-time graph

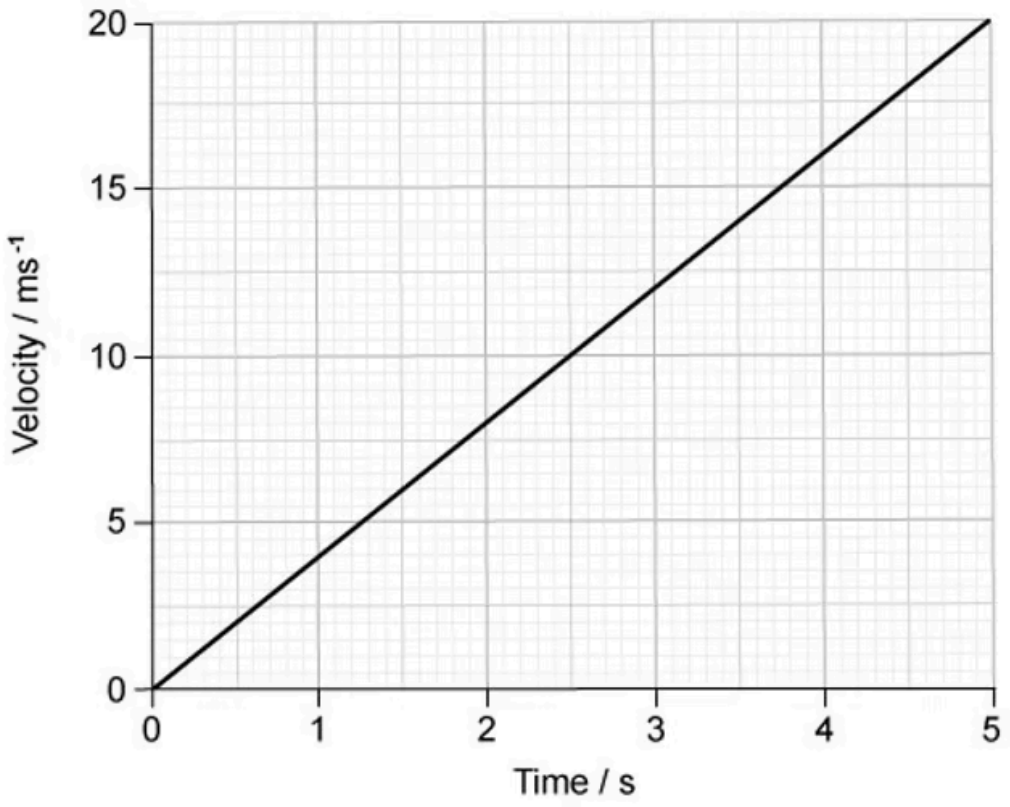
**B.** the area under a velocity-time graph

**C.** the slope of a displacement-time graph

**D.** the slope of a velocity-time graph

**(1 mark)**

**4** The velocity of an object during the first five seconds of its motion is shown on the graph




What is the displacement of the object in this time?

**A.** 4 m

**B.** 20 m

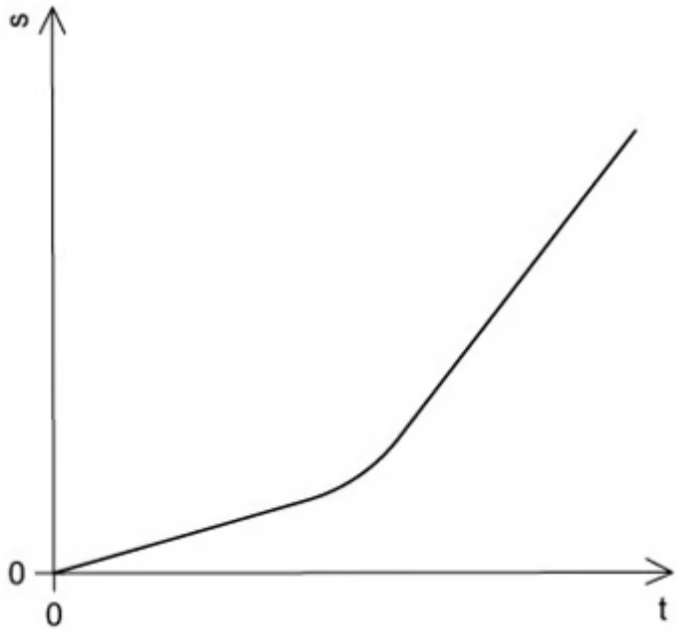
**C.** 50 m


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**D.** 100 m

**(1 mark)**

**5** The variation with time *t* of the displacement *s* moved by a body is shown below.




What can be deduced from the graph about the motion of the body?

**A.** it accelerates continuously

**B.** it starts from rest

**C.** the displacement is proportional to time

**D.** the speed changes

**(1 mark)**


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**6** An object is thrown with velocity 7.1 m $s^{-1}$ vertically upwards on the Moon. The acceleration due to gravity on the Moon is 1.62 m $s^{-2}$

What is the time taken for the object to return to its starting point?

**A.** 3.5 s

**B.** 4.4 s

**C.** 6.5 s

**D.** 8.8 s

**(1 mark)**

**7** A ball is thrown upwards with an initial vertical velocity of 2.2 m $s^{-1}$.

What is the maximum height reached by the ball?

**A.** 10 cm

**B.** 25 cm

**C.** 50 cm

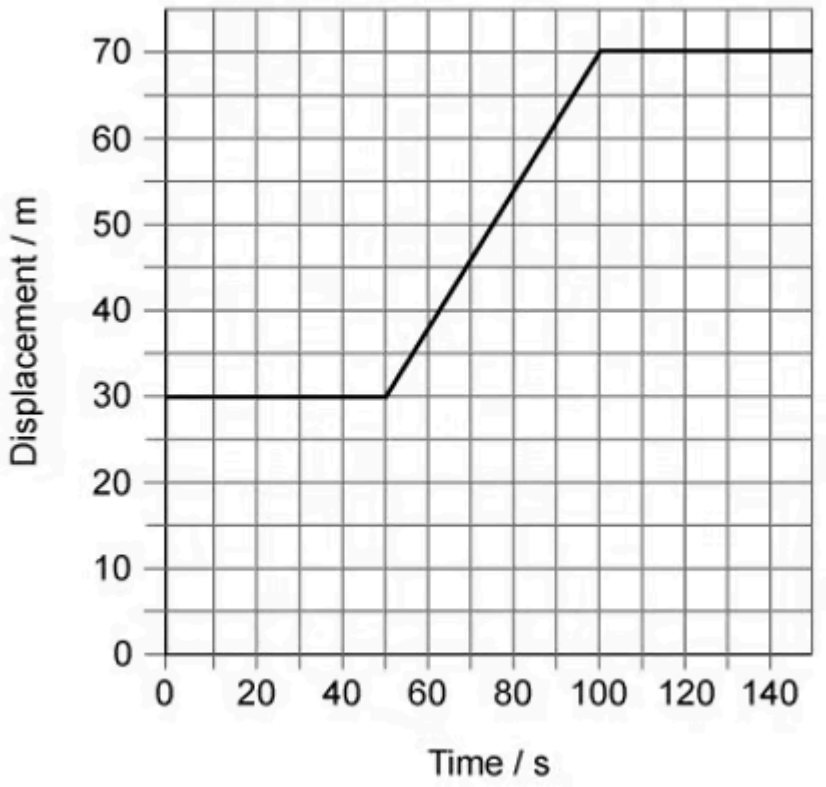
**D.** 75 cm

**(1 mark)**

**8** A car at rest in a traffic queue moves forward in a straight line and then comes to rest again.


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The graph shows how the car's displacement varies with time




What is its speed while it is moving?

**A.** 0.70 m $s^{-1}$

**B.** 0.80 m $s^{-1}$

**C.** 1.25 m $s^{-1}$

**D.** 1.40 m $s^{-1}$

**(1 mark)**


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**9** On a particular railway, a train driver applies the brake of the train at a yellow signal, a distance of 1.0 km from a red signal, where the train stops.

The maximum deceleration of the train is 1.1 m $s^{-2}$

Assuming uniform deceleration, what is the maximum safe speed of the train at the yellow signal?

**A.** 2.2 m $s^{-1}$

**B.** 20 m $s^{-1}$

**C.** 47 m $s^{-1}$

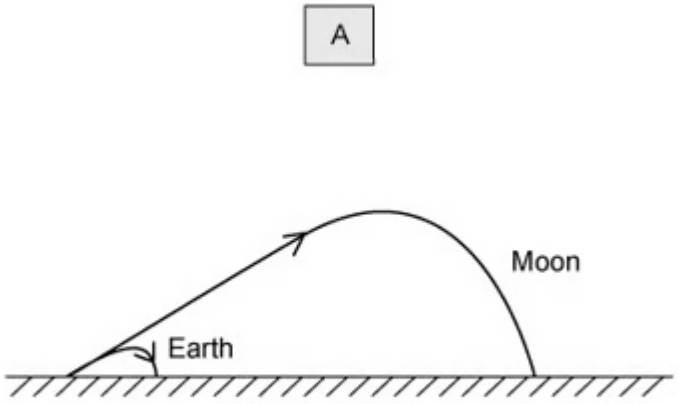
**D.** 220 m $s^{-1}$

**(1 mark)**

**10** A golf ball is hit with the same force and direction on the Earth and on the Moon.

Which diagram best represents the shapes of the paths taken by the golf ball?

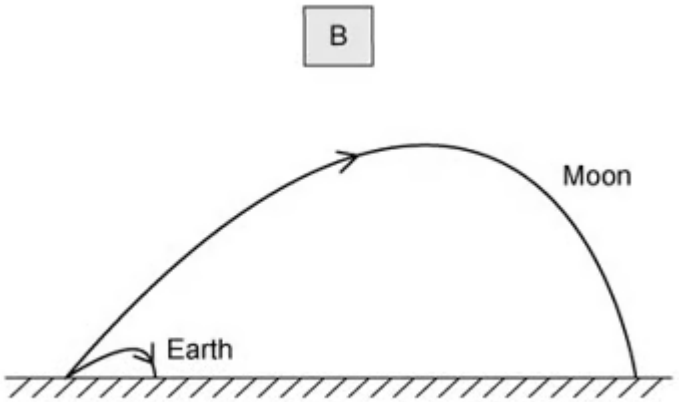
**A.**



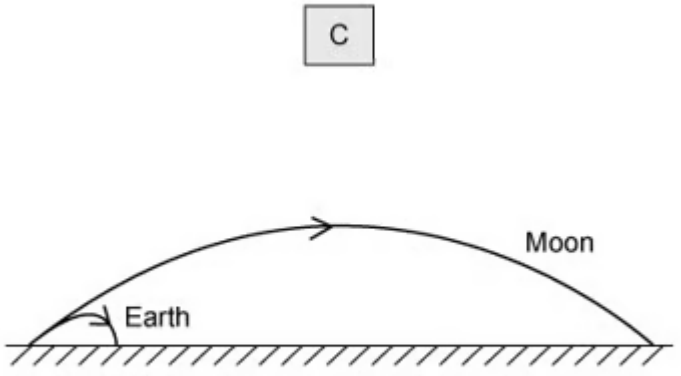


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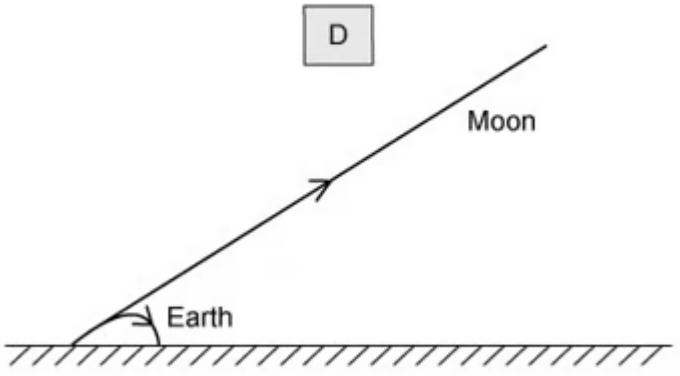
**B.**




**C.**




**D.**





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**(1 mark)**

**11** Which feature of a graph allows velocity to be determined?

**A.** the area under a displacement-time graph

**B.** the area under a velocity-time graph

**C.** the slope of a displacement-time graph

**D.** the slope of a velocity-time graph

**(1 mark)**

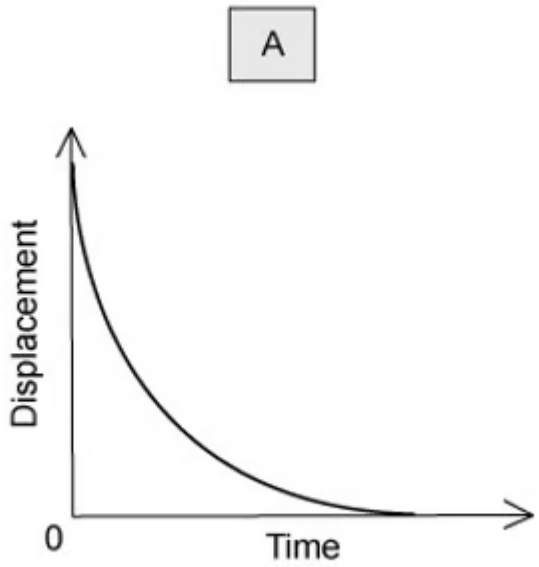

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# Medium Questions

**1** A sphere is released and falls. Its initial acceleration reduces until it eventually begins to travel at constant terminal velocity.

Which displacement-time graph best represents the motion of the sphere?

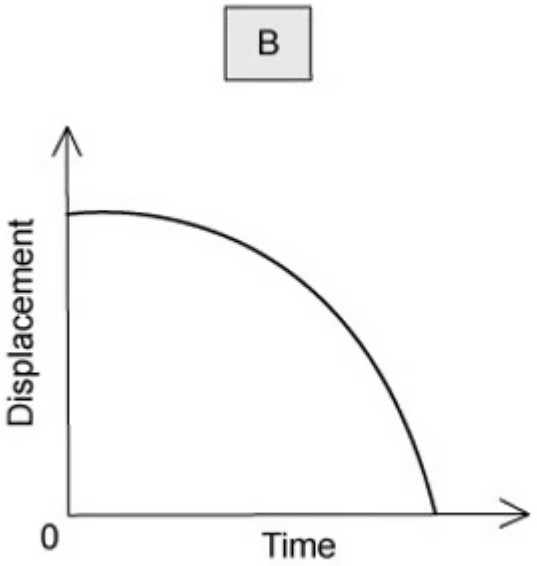
**A.**



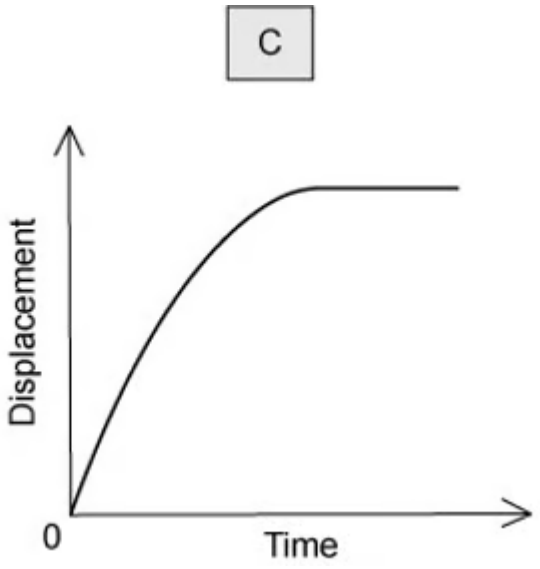


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**B.**



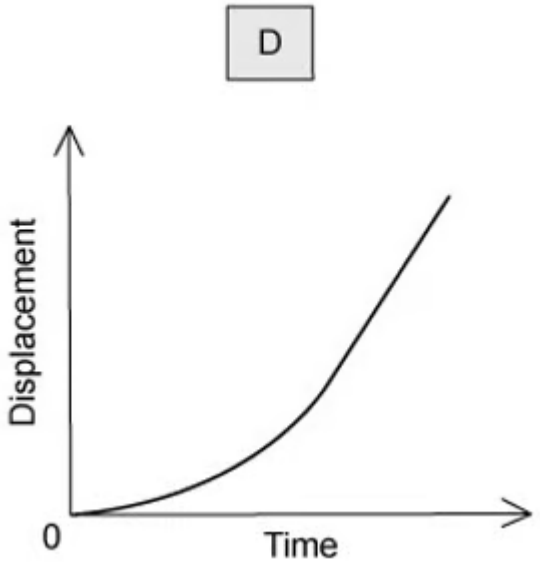

**C.**





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**D.**




**(1 mark)**

**2** A bullet is fired horizontally with speed *v* from a rifle. For a short time *t* after leaving the rifle, the only force affecting its motion is gravity. The acceleration of free fall is *g*.

Which expression gives the value of $\frac{\text{horizontal distance travelled in time } t}{\text{vertical distance travelled in time } t}$ ?

**A.** $\frac{vt}{g}$

**B.** $\frac{v}{gt}$

**C.** $\frac{2vt}{g}$

**D.** $\frac{2v}{gt}$

**(1 mark)**

**3** The acceleration of free fall on the Moon is one-sixth of that on Earth. On Earth it takes time *t* for a stone to fall from rest a distance of 2 m.

What is the time taken for a stone to fall from rest a distance of 2 m on the Moon?


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**A.** $6t$

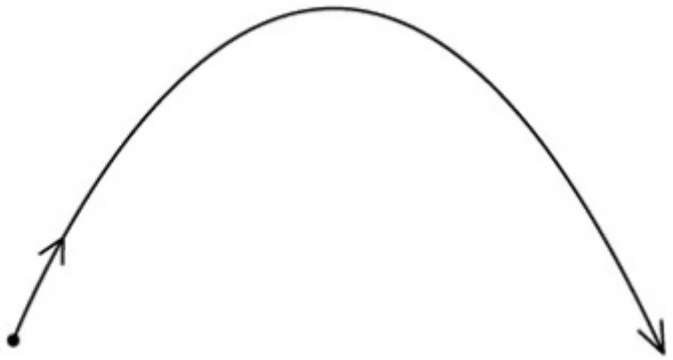
**B.** $\frac{t}{6}$

**C.** $t\sqrt{6}$

**D.** $\frac{t}{\sqrt{6}}$

**(1 mark)**

**4** A stone is thrown upwards and follows a curved path.



Air resistance is negligible.

Why does the path have this shape?

**A.** the stone has a constant horizontal velocity and constant vertical acceleration

**B.** the stone has a constant horizontal acceleration and constant vertical velocity

**C.** the stone has a constant upward acceleration followed by a constant downward acceleration

**D.** the stone has a constant upward velocity followed by a constant downward velocity

**(1 mark)**


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**5** A train passes through a station at a constant speed of 20 m $s^{-1}$

A second train is at rest at the station. It begins to leave the station with a uniform acceleration of 1.0 m $s^{-2}$ just as the first train goes past.

How much time passes before the second train overtakes the first train?

**A.** 5 s

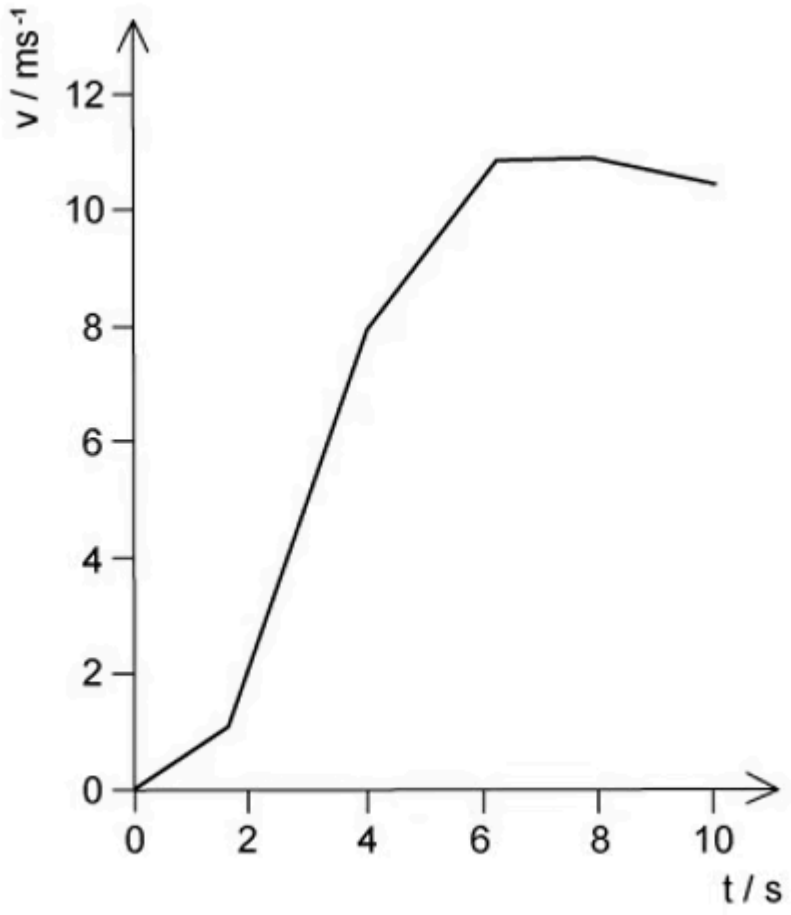
**B.** 10 s

**C.** 20 s

**D.** 40 s

**(1 mark)**

**6** The graph shows how the speed *v* of a sprinter changes with time *t* during a 100 m race.




What is the best estimate of the maximum acceleration of the sprinter?

**A.** 3 m $s^{-2}$

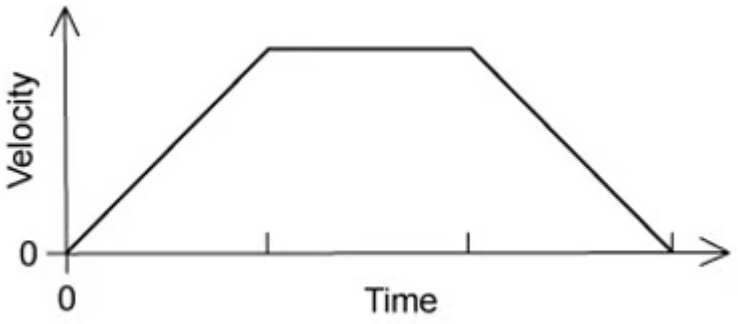
**B.** 6 m $s^{-2}$

**C.** 9 m $s^{-2}$

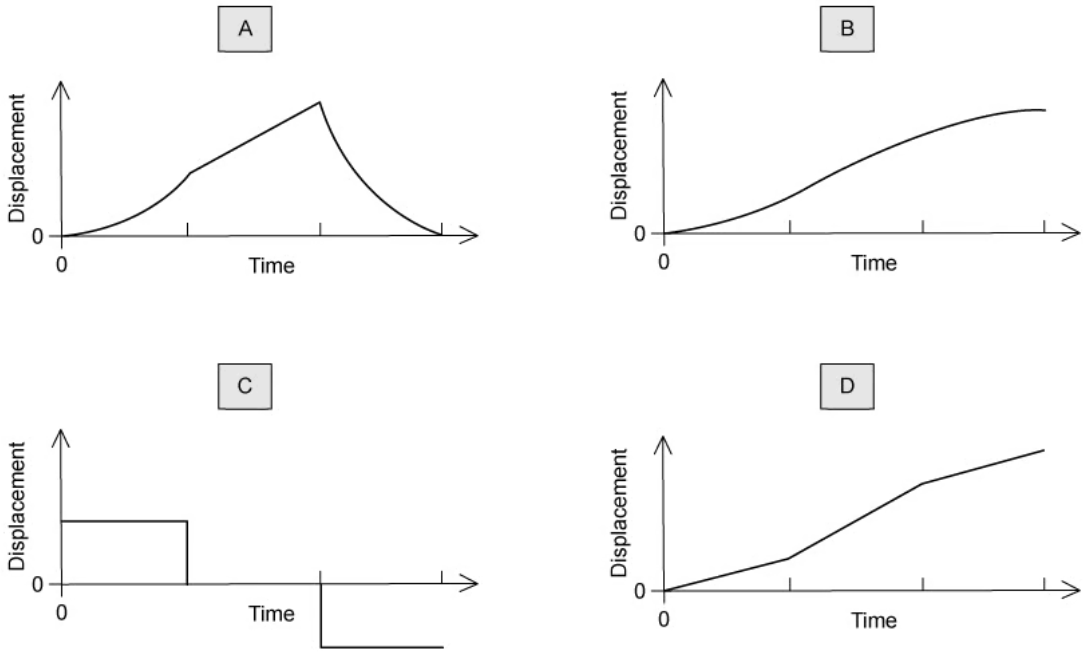
**D.** 12 m $s^{-2}$

**(1 mark)**

**7** The graph of velocity against time for an object moving in a straight line is shown




What is the corresponding graph of displacement against time?



**(1 mark)**

**8** In a cathode-ray tube, an electron is accelerated uniformly in a straight line from a speed of $4 \times 10^5$ m s$^{-1}$ to one tenth of the speed of light over a distance of 10 mm

What is the acceleration of the electron?

**A.** $4.5 \times 10^3$ m s$^{-2}$

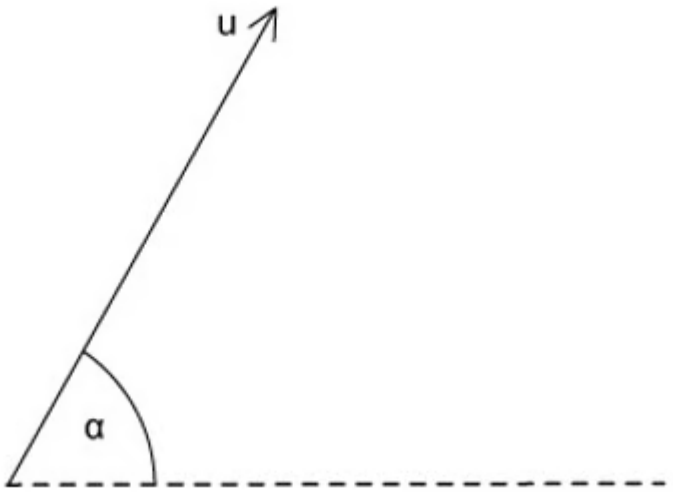
**B.** $4.5 \times 10^6$ m s$^{-2}$

**C.** $4.5 \times 10^{13}$ m s$^{-2}$

**D.** $4.5 \times 10^{16}$ m s$^{-2}$

**(1 mark)**

**9** A projectile is fired at an angle $\alpha$ to the horizontal at a speed $u$, as shown.




What are the vertical and horizontal components of its velocity after a time $t$?

Assume that air resistance is negligible. The acceleration of free fall is $g$.

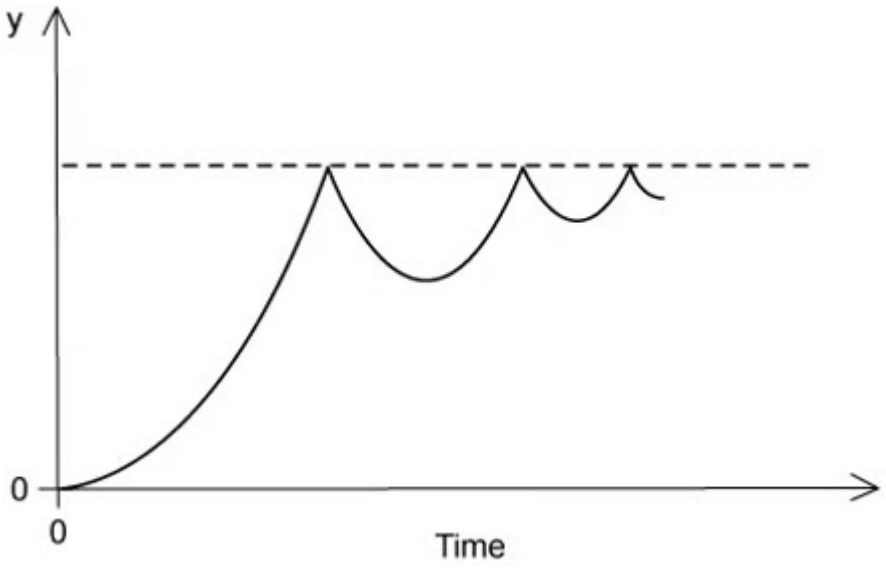

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| | Vertical component | Horizontal component |
|---|---|---|
| **A** | $u \sin \alpha$ | $u \cos \alpha$ |
| **B** | $u \sin \alpha - gt$ | $u \cos \alpha - gt$ |
| **C** | $u \sin \alpha - gt$ | $u \cos \alpha$ |
| **D** | $u \cos \alpha$ | $u \sin \alpha - gt$ |

**(1 mark)**

**10** A ball is released from rest above a horizontal surface and bounces several times.

The graph shows how, for this ball, a quantity *y* varies with time.




What is the quantity *y*?

**A.** acceleration

**B.** displacement

**C.** kinetic energy

**D.** velocity

**(1 mark)**

**11** A spacecraft is launched from Earth's surface and accelerates uniformly in a straight line from an initial speed of $2 \times 10^3 \text{ m s}^{-1}$ to a final speed of $6 \times 10^3 \text{ m s}^{-1}$ over a distance of 400 km.

What is the acceleration of the spacecraft?

**A.** $4 \times 10^{-3} \text{ m s}^{-2}$

**B.** $4 \times 10^{1} \text{ m s}^{-2}$

**C.** $4 \times 10^{3} \text{ m s}^{-2}$

**D.** $4 \times 10^{6} \text{ m s}^{-2}$

**(1 mark)**


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# Hard Questions

**1** A body having uniform acceleration *a* increases its velocity from *u* to *v* in time *t*.

Which expression would **not** give a correct value for the body's displacement during time *t* ?

**A.** $ut + \frac{1}{2}at^2$

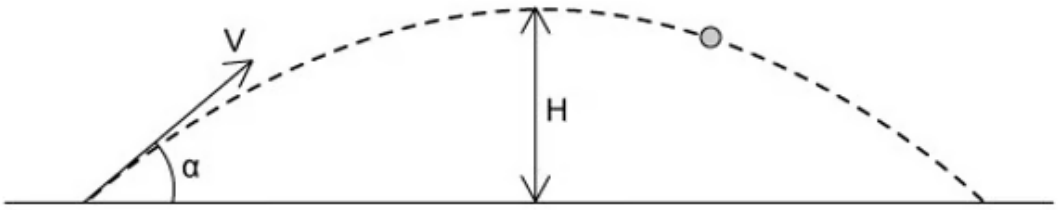
**B.** $vt - \frac{1}{2}at^2$

**C.** $\frac{(v+u)(v-u)}{2a}$

**D.** $\frac{(v-u)t}{2}$

**(1 mark)**

**2** A cannon fires a cannonball with an initial speed *v* at an angle α to the horizontal.




Which equation is correct for the maximum height H reached?

**A.** $H = \frac{v \sin \alpha}{2g}$

**B.** $H = \frac{g \sin \alpha}{2v}$

**C.** $H = \frac{(v \sin \alpha)^2}{2g}$


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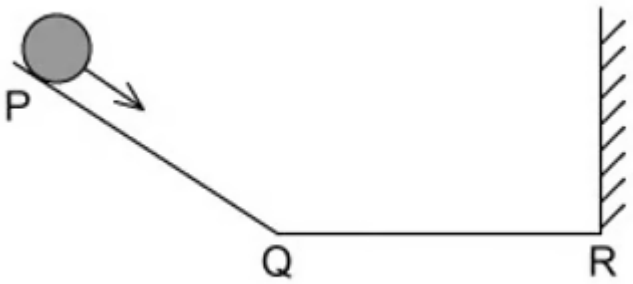
**D.** $H = \frac{g^2 \sin \alpha}{2v}$

**(1 mark)**

**3** A ball is released from rest on a smooth slope PQR.

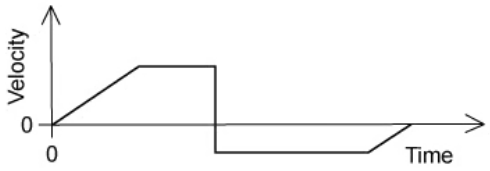
It moves down the slope, along a smooth horizontal surface QR and rebounds inelastically at R.

Then it moves back to Q and comes to rest momentarily somewhere on PQ.

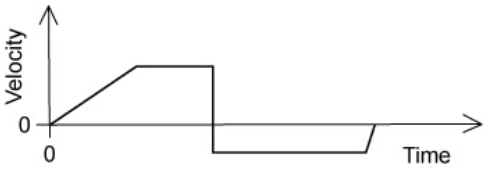



Which velocity-time graph represents the motion of the ball?

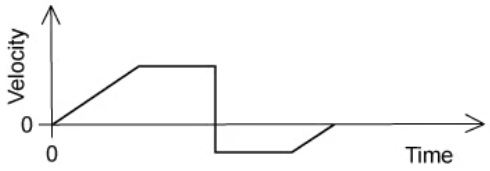
A



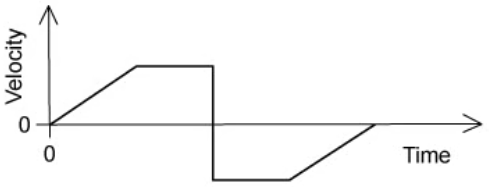

B




C

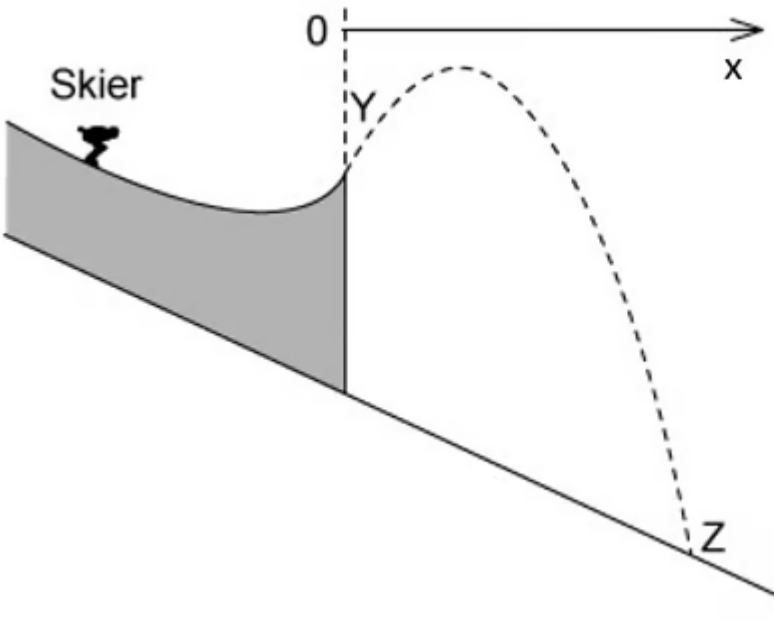



D

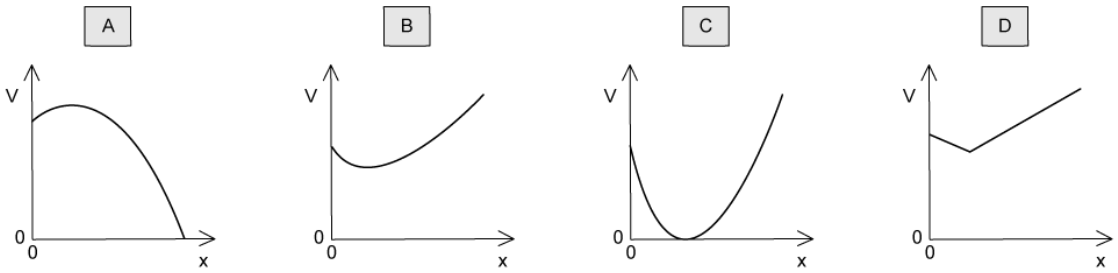


**(1 mark)**

**4** The dotted line shows the path of a competitor in a ski-jumping competition.



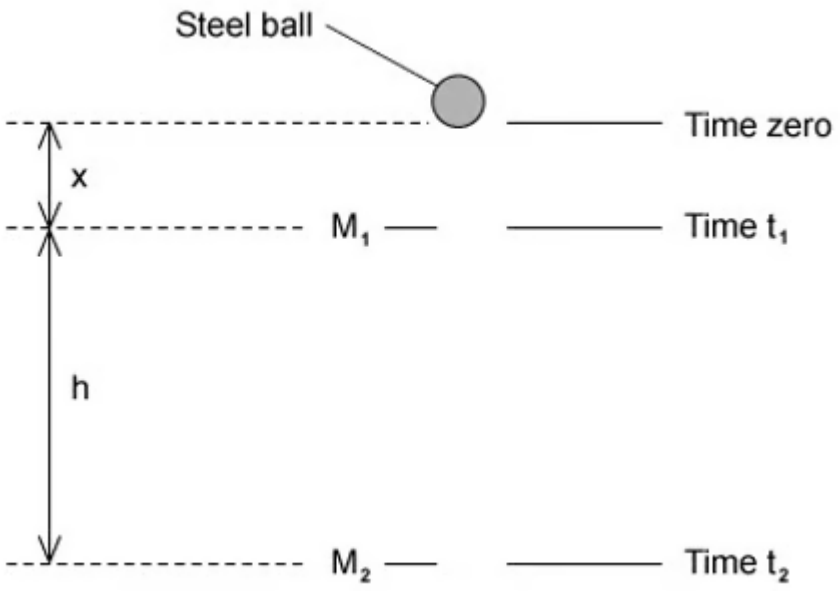

Ignoring air resistance, which graph best represents the variation of his speed *v* with the horizontal distance *x* covered from the start of his jump at Y before landing at Z?




**(1 mark)**


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**5** Two markers $M_1$ and $M_2$ are set up a vertical distance $h$ apart.




A steel ball is released at time zero from a point a distance $x$ above $M_1$. The ball reaches $M_1$ at time $t_1$ and reaches $M_2$ at time $t_2$. The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

**A.** $\frac{2h}{t_2^2}$

**B.** $\frac{2h}{(t_1 + t_2)}$

**C.** $\frac{2h}{t_2^2 - t_1^2}$

**D.** $\frac{2h}{(t_2 - t_1)^2}$

**(1 mark)**


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**6** A sprinter runs a 200 m race in a straight line. He starts from rest and accelerates at 2.5 m $s^{-2}$ until reaching a maximum speed of 10 m $s^{-1}$. He maintains this speed until he crosses the finish line.

How long does it take the sprinter to run the race?

**A.** 8 s

**B.** 20 s

**C.** 22 s

**D.** 40 s

**(1 mark)**

**7** In order that a train can stop safely, it will always pass a signal showing a yellow light before it reaches a signal showing a red light.

Drivers apply the brake at the yellow light and this results in a uniform deceleration to stop exactly at the red light. The distance between the red and yellow lights is $x$.

What must be the minimum distance between the lights if the train speed is increased by 25 %, without changing the deceleration of the trains?

**A.** 1.20$x$

**B.** 1.25$x$

**C.** 1.44$x$

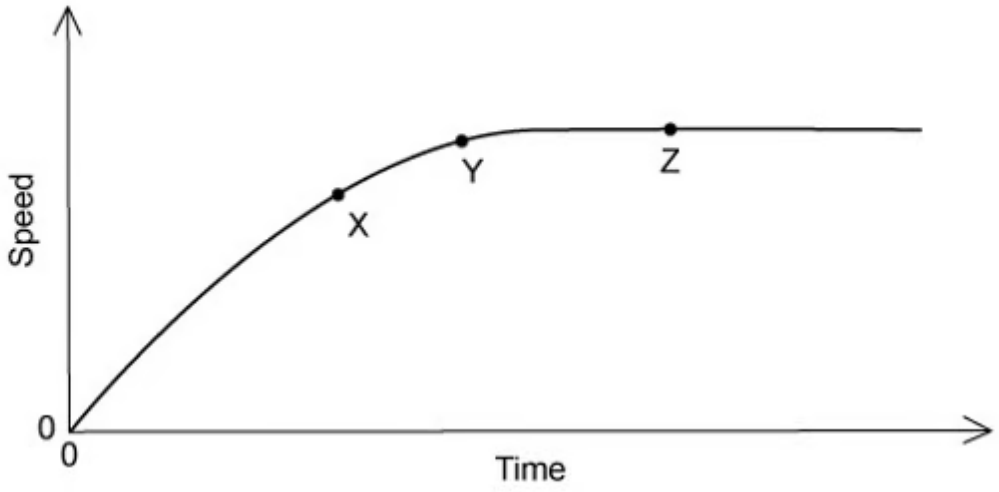
**D.** 1.56$x$

**(1 mark)**

**8** A raindrop falls vertically from rest in air. The variation with time of the speed of the


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raindrop is shown in the graph.




Which statement about the raindrop is correct?

**A.** at point X, the raindrop has an acceleration of 9.81 m $s^{-2}$

**B.** at point Z, the force on the raindrop due to air resistance has reached its maximum value and so the acceleration of the raindrop has also reached its maximum value

**C.** at point Z, the force due to air resistance is equal and opposite to the weight of the raindrop and so the speed of the raindrop is zero

**D.** the resultant force on the raindrop at point Y is less than the resultant force on the raindrop at point X

**(1 mark)**

**9** An aeroplane travels at an average speed of 700 km $h^{-1}$ on an outward flight and at 300 km $h^{-1}$ on the return flight over the same distance.

What is the average speed of the whole flight?

**A.** 400 km $h^{-1}$

**B.** 420 km $h^{-1}$

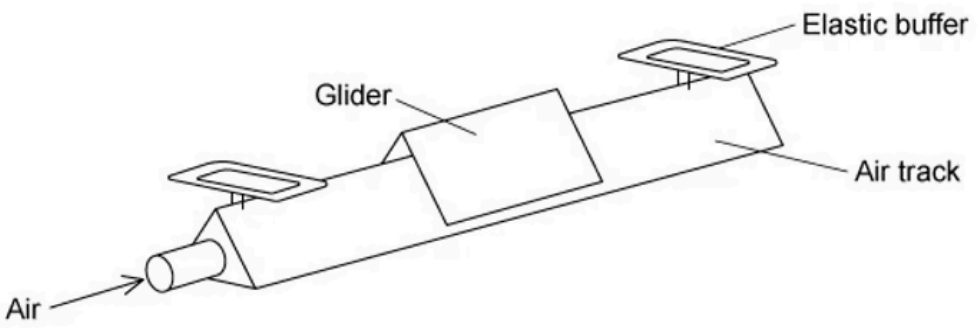
**C.** 480 km $h^{-1}$

**D.** 500 km $h^{-1}$

**(1 mark)**


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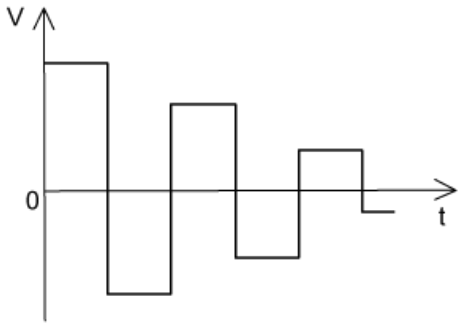
**10** A small glider moves along a friction-free horizontal air track as shown below.



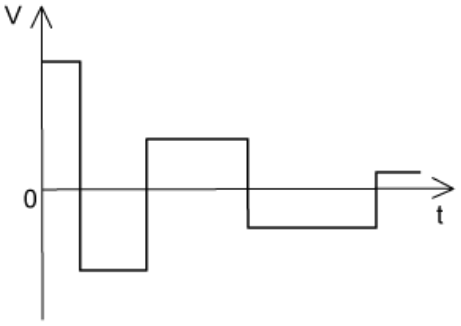

At each end of the air track there is a perfectly elastic buffer.

Which graph represents the variation with time *t* of the velocity *v* of the glider as it moves between the two buffers?

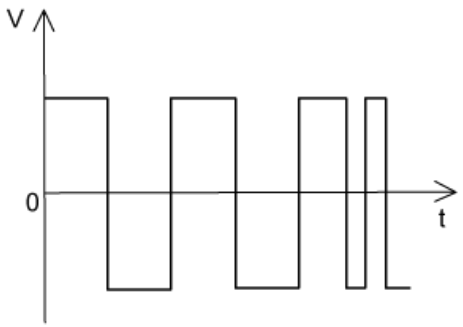
A



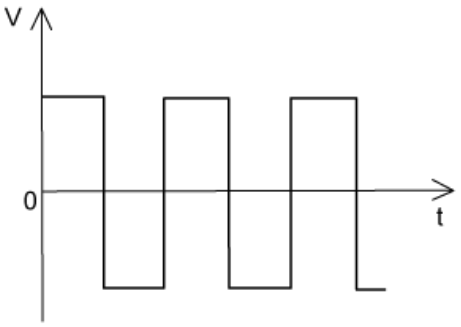

B




C




D



**(1 mark)**